

EXAMEN COMPLEXIVO PRÁCTICO

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Cohorte 2024-2
Ciclo 2026-1

Indicaciones Generales

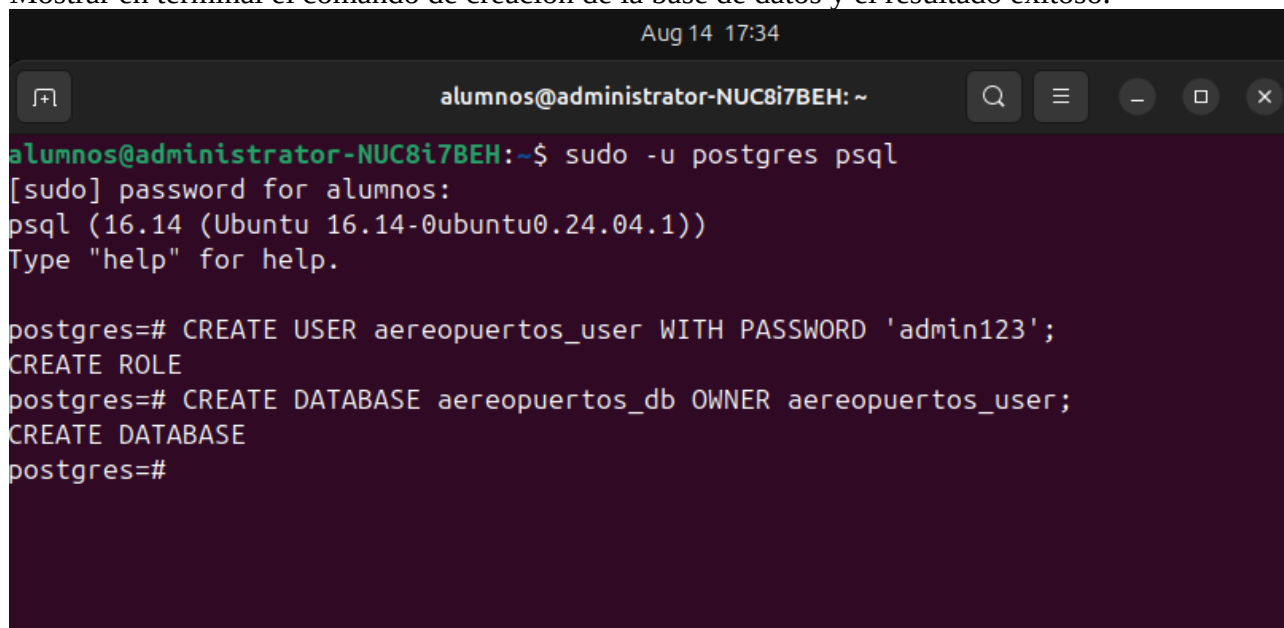
Todas las capturas deben mostrar:

- Fecha y hora visible del sistema.
- Terminal o navegador completo (sin recortes parciales).
- Las capturas deben estar numeradas consecutivamente.
- Deben entregarse en un único documento PDF o Word ordenado según esta plantilla.
- No se aceptarán imágenes editadas o fragmentadas.

SECCIÓN 1: BASE DE DATOS RELACIONAL (POSTGRESQL)

Captura 1 – Creación de Base de Datos

Mostrar en terminal el comando de creación de la base de datos y el resultado exitoso.



```
Aug 14 17:34
alumnos@administrator-NUC8i7BEH: ~
alumnos@administrator-NUC8i7BEH:~$ sudo -u postgres psql
[sudo] password for alumnos:
psql (16.14 (Ubuntu 16.14-0ubuntu0.24.04.1))
Type "help" for help.

postgres=# CREATE USER aereopuertos_user WITH PASSWORD 'admin123';
CREATE ROLE
postgres=# CREATE DATABASE aereopuertos_db OWNER aereopuertos_user;
CREATE DATABASE
postgres=#
```

Captura 2 – Creación de Usuario y Asignación de Permisos

Mostrar los comandos utilizados para crear el usuario y asignar permisos mínimos necesarios.

```
Aug 14 17:36
alumnos@administrator-NUC8i7BEH: ~
alumnos@administrator-NUC8i7BEH:~$ sudo -u postgres psql
[sudo] password for alumnos:
psql (16.14 (Ubuntu 16.14-0ubuntu0.24.04.1))
Type "help" for help.

postgres=# CREATE USER aereopuertos_user WITH PASSWORD 'admin123';
CREATE ROLE
postgres=# CREATE DATABASE aereopuertos_db OWNER aereopuertos_user;
CREATE DATABASE
postgres=# \c aereopuertos_db
You are now connected to database "aereopuertos_db" as user "postgres".
aereopuertos_db=# ALTER SCHEMA public OWNER TO aereopuertos_user;
ALTER SCHEMA
aereopuertos_db=# GRANT ALL ON SCHEMA public TO aereopuertos_user;
GRANT
aereopuertos_db=# GRANT CREATE ON SCHEMA public TO aereopuertos_user;
GRANT
aereopuertos_db=# ALTER DEFAULT PRIVILEGES FOR USER aereopuertos_user IN SCHEMA
public
GRANT ALL ON TABLES TO aereopuertos_user;
ALTER DEFAULT PRIVILEGES
aereopuertos_db=# ALTER DEFAULT PRIVILEGES FOR USER aereopuertos_user IN SCHEMA
public
GRANT ALL ON SEQUENCES TO aereopuertos_user;
ALTER DEFAULT PRIVILEGES
aereopuertos_db=# ALTER DEFAULT PRIVILEGES FOR USER aereopuertos_user IN SCHEMA
public
GRANT ALL ON FUNCTIONS TO aereopuertos_user;
ALTER DEFAULT PRIVILEGES
aereopuertos_db=#
```

Captura 3 – Conexión con el Usuario Creado

Mostrar conexión exitosa mediante psql y listado de bases de datos.

```

Aug 14 18:05
alumnos@administrator-NUC8i7BEH: ~
alumnos@administrator-NUC8i7BEH: ~
alumnos@administrator-NUC8i7BEH: ~

List of databases
+-----+-----+-----+-----+-----+
| Name          | Owner          | Encoding | Locale Provider | Collate |
| Ctype        | ICU Locale     | ICU Rules | Access privileges |         |
+-----+-----+-----+-----+-----+
| aereopuertos_db | aereopuertos_user | UTF8     | libc            | en_US.UTF-8 |
| en_US.UTF-8 | | | | |
| postgres        | postgres       | UTF8     | libc            | en_US.UTF-8 |
| en_US.UTF-8 | | | | |
| template0       | postgres       | UTF8     | libc            | en_US.UTF-8 |
| en_US.UTF-8 | | | | =c/postgres +
| | | | postgres=CTc/postgres
| template1       | postgres       | UTF8     | libc            | en_US.UTF-8 |
| en_US.UTF-8 | | | | =c/postgres +
| | | | postgres=CTc/postgres
(4 rows)

(END)

```

Captura 4 – Tablas Generadas por Migración

Mostrar el listado de tablas creadas mediante migraciones (comando \dt).

```

Aug 14 18:09
alumnos@administrator-NUC8i7BEH: ~
alumnos@administrator-NUC8i7BEH: ~
alumnos@administrator-NUC8i7BEH: ~

aereopuertos_db-> \dt
                                List of relations
 Schema |          Name          | Type |      Owner
-----+-----+-----+-----
 public | auth_group             | table | aereopuertos_user
 public | auth_group_permissions | table | aereopuertos_user
 public | auth_permission        | table | aereopuertos_user
 public | auth_user              | table | aereopuertos_user
 public | auth_user_groups       | table | aereopuertos_user
 public | auth_user_user_permissions | table | aereopuertos_user
 public | django_admin_log       | table | aereopuertos_user
 public | django_content_type    | table | aereopuertos_user
 public | django_migrations      | table | aereopuertos_user
 public | django_session         | table | aereopuertos_user
 public | flights                | table | aereopuertos_user
 public | gates                  | table | aereopuertos_user
(12 rows)

aereopuertos_db->

```

Captura 5 – Estructura de Tablas

Mostrar la estructura detallada de las dos tablas principales (comando \d nombre_tabla), donde se evidencien tipos de datos.



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```
Aug 14 18:10
alumnos@administrator-NUC8i7BEH: ~
alumnos@administrator-NUC8i7BEH: ~
alumnos@administrator-NUC8i7BEH: ~

Table "public.gates"
Column | Type | Collation | Nullable | Default
-----+-----+-----+-----+-----
id | bigint | | not null | generated by default as identity
code | character varying(10) | | not null |
terminal | character varying(20) | | not null |
is_available | boolean | | not null |
created_at | timestamp with time zone | | not null |

Indexes:
    "gates_pkey" PRIMARY KEY, btree (id)
    "gates_code_514a9f49_like" btree (code varchar_pattern_ops)
    "gates_code_key" UNIQUE CONSTRAINT, btree (code)

Referenced by:
    TABLE "flights" CONSTRAINT "flights_gate_id_8a7976c1_fk_gates_id" FOREIGN KEY (gate_id) REFERENCES gates(id) DEFERRABLE INITIALLY DEFERRED

:
```



```

Aug 14 18:10
alumnos@administrator-NUC8i7BEH: ~
alumnos@administrator-NUC8i7BEH: ~
alumnos@administrator-NUC8i7BEH: ~

Table "public.flights"
Column | Type | Collation | Nullable |
Default
-----+-----+-----+-----+-----
id | bigint | | not null | generated by
default as identity
flight_number | character varying(20) | | not null |
destination | character varying(100) | | not null |
status | character varying(20) | | not null |
departure_time | timestamp with time zone | | not null |
created_at | timestamp with time zone | | not null |
gate_id | bigint | | not null |
Indexes:
    "flights_pkey" PRIMARY KEY, btree (id)
    "flights_gate_id_8a7976c1" btree (gate_id)
Foreign-key constraints:
    "flights_gate_id_8a7976c1_fk_gates_id" FOREIGN KEY (gate_id) REFERENCES gate
s(id) DEFERRABLE INITIALLY DEFERRED
:

```

Captura 6 – Creación de Índice

Mostrar el comando de creación del índice y la verificación de su existencia.

```

Aug 14 18:15
alumnos@administrator-NUC8i7BEH: ~
alumnos@administrator-NUC8i7BEH: ~
alumnos@administrator-NUC8i7BEH: ~

aereopuertos_db=> CREATE INDEX idx_flights_status ON flights(status);
CREATE INDEX
aereopuertos_db=> EXPLAIN SELECT * FROM flights WHERE status = 'SCHEDULED';
aereopuertos_db=> SELECT * FROM flights;
aereopuertos_db=>

```



```

Aug 14 18:16
alumnos@administrator-NUC8i7BEH: ~
QUERY PLAN
-----
Index Scan using idx_flights_status on flights (cost=0.14..8.16 rows=1 width=366)
Index Cond: ((status)::text = 'SCHEDULED'::text)
(2 rows)

~
(END)

```

```

Aug 14 18:16
alumnos@administrator-NUC8i7BEH: ~
id | flight_number | destination | status | departure_time | created_at | gate_id
---+-----+-----+-----+-----+-----+-----
(0 rows)

(END)

```

Captura 7 – Creación de Vista

Mostrar el comando de creación de la vista y una consulta ejecutada sobre ella.

```
Aug 14 18:17
alumnos@administrator-NUC8i7BEH: ~
alumnos@administrator-NUC8i7BEH: ~
aereopuertos_db=> CREATE VIEW vw_active_flights AS
SELECT * FROM flights
WHERE status IN ('SCHEDULED', 'BOARDING');
CREATE VIEW
aereopuertos_db=> SELECT * FROM vw_active_flights;
aereopuertos_db=>
```

```
Aug 14 18:17
alumnos@administrator-NUC8i7BEH: ~
alumnos@administrator-NUC8i7BEH: ~
id | flight_number | destination | status | departure_time | created_at | gate_
id
-----+-----+-----+-----+-----+-----+-----
(0 rows)

~
~
(END)
```

Captura 8 – Función o Trigger

Mostrar la creación y prueba funcional de la función o trigger implementado.



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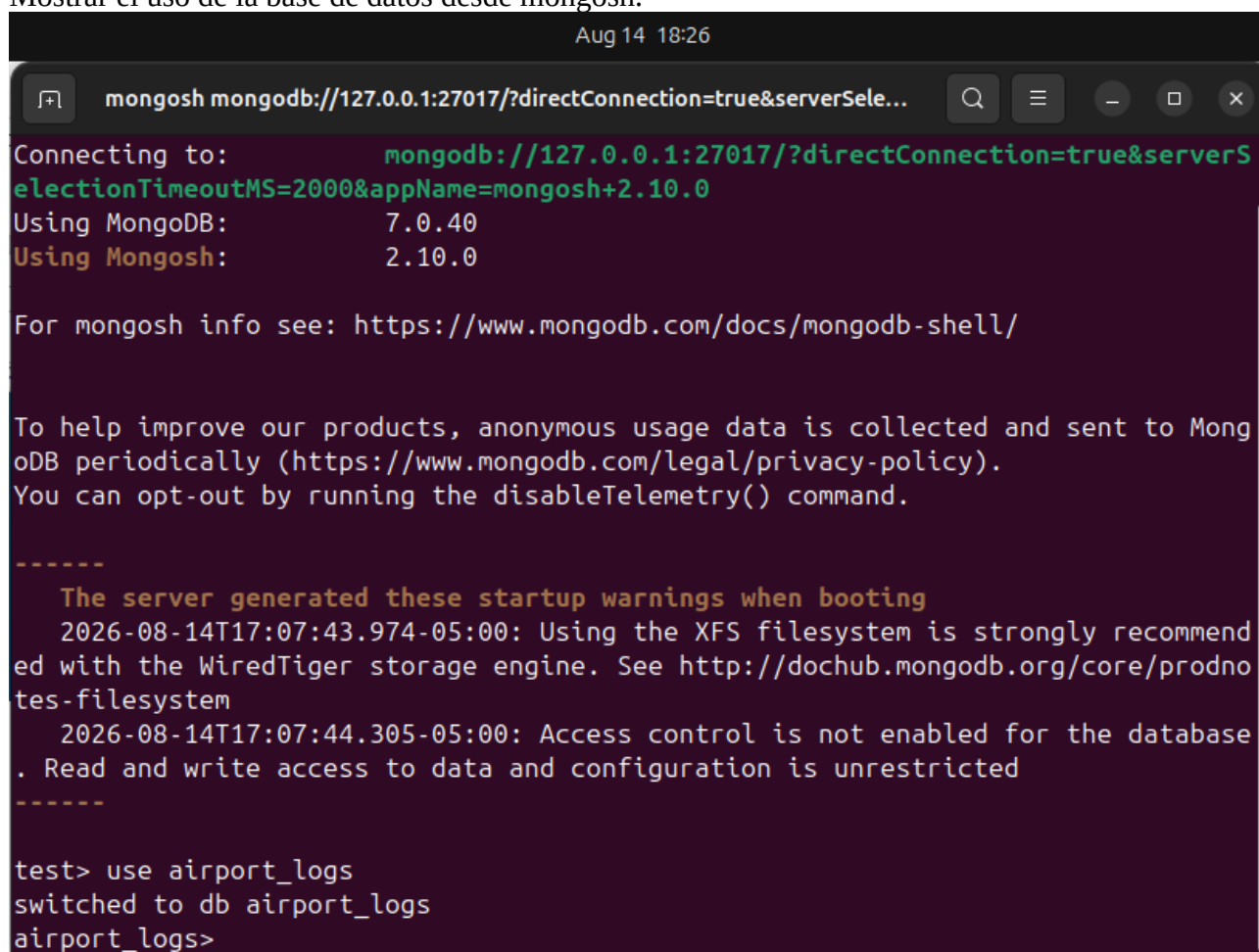
```
Aug 14 18:19
alumnos@administrator-NUC8i7BEH: ~
alumnos@administrator-NUC8i7BEH: ~
aereopuertos_db=> CREATE OR REPLACE FUNCTION check_flight_departure_time()
RETURNS TRIGGER AS $$
BEGIN
    IF NEW.departure_time < NEW.created_at THEN
        RAISE EXCEPTION 'La fecha de salida (departure_time) no puede ser anteri
or a la fecha de creación (created_at)';
    END IF;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;
CREATE FUNCTION
aereopuertos_db=> CREATE TRIGGER trg_check_flight_departure_time
BEFORE INSERT OR UPDATE ON flights
FOR EACH ROW
EXECUTE FUNCTION check_flight_departure_time();
CREATE TRIGGER
aereopuertos_db=>
```



SECCIÓN 2: BASE DE DATOS NO RELACIONAL (MONGODB)

Captura 9 – Creación y Selección de Base de Datos

Mostrar el uso de la base de datos desde mongosh.



```
Aug 14 18:26
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSele...
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverS
electionTimeoutMS=2000&appName=mongosh+2.10.0
Using MongoDB:      7.0.40
Using Mongosh:      2.10.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

To help improve our products, anonymous usage data is collected and sent to Mong
oDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

-----
The server generated these startup warnings when booting
2026-08-14T17:07:43.974-05:00: Using the XFS filesystem is strongly recommend
ed with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodno
tes-filesystem
2026-08-14T17:07:44.305-05:00: Access control is not enabled for the database
. Read and write access to data and configuration is unrestricted
-----

test> use airport_logs
switched to db airport_logs
airport_logs>
```

Captura 10 – Creación de Usuario

Mostrar el comando de creación del usuario con roles asignados.

```
Aug 14 18:28
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSele...
airport_logs> db.createUser({
|   user: "mongo_backend_user",
|   pwd: "exa_2026_ute",
|   roles: [{ role: "readWrite", db: "airport_logs" }]
| })
{ ok: 1 }
airport_logs> db.auth("mongo_backend_user", "exa_2026_ute")
{ ok: 1 }
airport_logs>
```

Captura 11 – Creación o Verificación de Colecciones

Mostrar las colecciones existentes y la inserción de un documento de prueba.

```
Aug 14 18:29
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSele...
airport_logs> db.createCollection("airlines")
{ ok: 1 }
airport_logs> db.createCollection("flight_events")
{ ok: 1 }
airport_logs> show collections
airlines
flight_events
airport_logs>
```

Captura 12 – Creación de Índice

Mostrar el comando createIndex() y la verificación con getIndexes().


```
Aug 14 18:31
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSele...
airport_logs> db.flight_events.createIndex({ flight_id: 1 })
flight_id_1
airport_logs> db.flight_events.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { flight_id: 1 }, name: 'flight_id_1' }
]
airport_logs>
```

Captura 13 – Consulta por Identificador

Mostrar una consulta filtrando por el identificador relacionado con la tabla relacional.

```
Aug 14 18:37
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSele...
airport_logs> db.flight_events.find({ flight_id: 1 })
[
  {
    _id: ObjectId('6a7fa697e278e89afcc55fb2'),
    flight_id: 1,
    event_type: 'CREATED',
    source: 'SYSTEM',
    note: 'Vuelo de prueba',
    created_at: ISODate('2026-08-14T23:36:55.088Z')
  }
]
airport_logs>
```

Captura 14 – Consulta por Rango de Fechas

Mostrar una consulta utilizando el campo de fecha.



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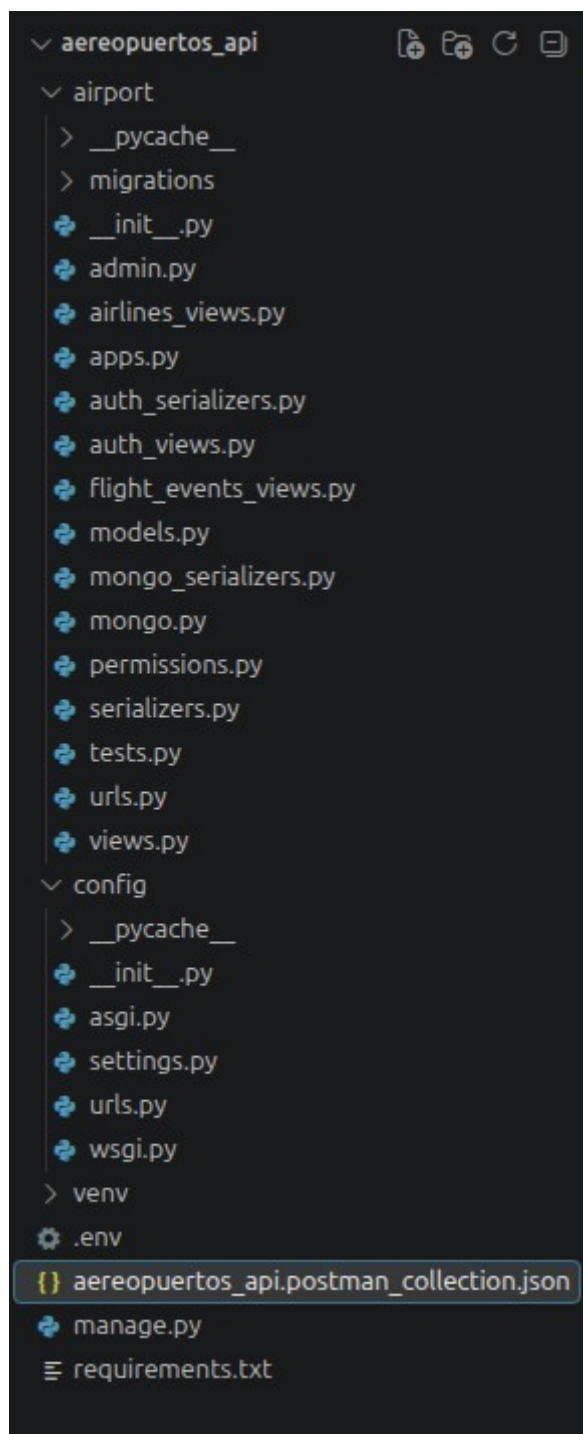
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```
Aug 14 18:37
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSele...
airport_logs> db.flight_events.find({
|   created_at: {
|     $gte: ISODate("2026-01-01T00:00:00Z"),
|     $lte: ISODate("2026-12-31T23:59:59Z")
|   }
| })
[
  {
    _id: ObjectId('6a7fa697e278e89afcc55fb2'),
    flight_id: 1,
    event_type: 'CREATED',
    source: 'SYSTEM',
    note: 'Vuelo de prueba',
    created_at: ISODate('2026-08-14T23:36:55.088Z')
  }
]
airport_logs>
```

SECCIÓN 3: BACKEND – DJANGO REST

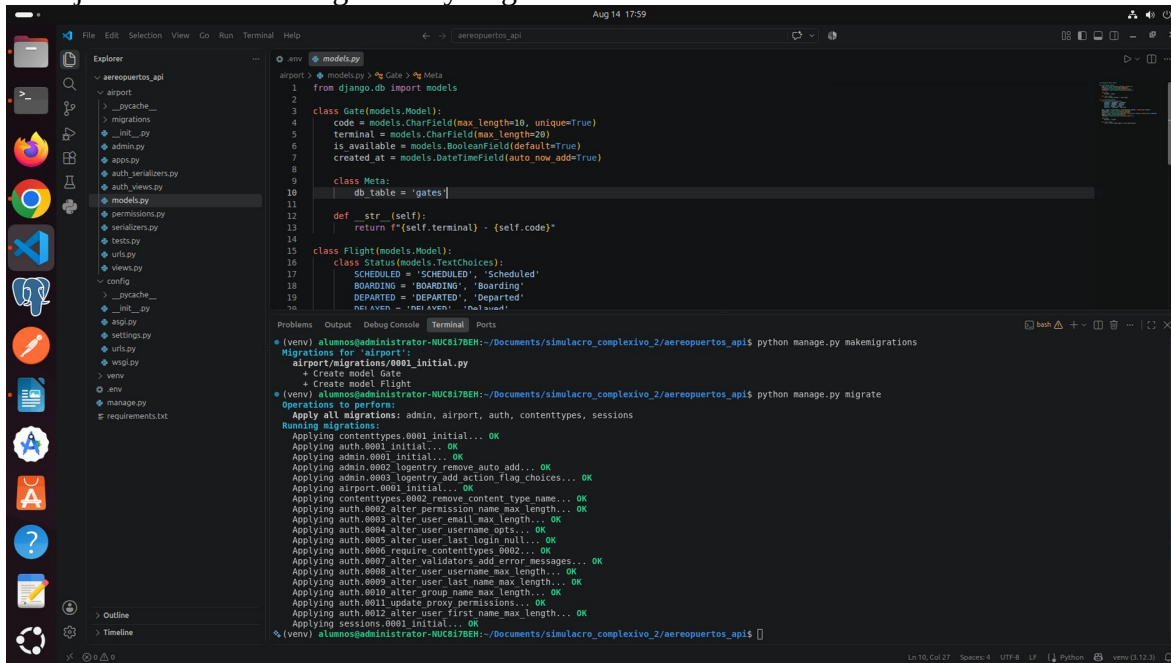
Captura 15 – Creación del Proyecto y Aplicación

Mostrar estructura del proyecto y aplicación creada.



Captura 16 – Migraciones Ejecutadas

Mostrar ejecución de makemigrations y migrate.



```

1 from django.db import models
2
3 class Gate(models.Model):
4     code = models.CharField(max_length=10, unique=True)
5     terminal = models.CharField(max_length=20)
6     is_available = models.BooleanField(default=True)
7     created_at = models.DateTimeField(auto_now_add=True)
8
9     class Meta:
10         db_table = 'gates'
11
12     def __str__(self):
13         return f'{self.terminal} - {self.code}'
14
15 class Flight(models.Model):
16     class Status(models.TextChoices):
17         SCHEDULED = 'SCHEDULED', 'Scheduled'
18         BOARDING = 'BOARDING', 'Boarding'
19         DEPARTED = 'DEPARTED', 'Departed'
20         NOT_AVAILABLE = 'NOT_AVAILABLE', 'Not Available'

```

(venv) alumnos@administrator-NUC817BEN:~/Documents/simulacro_complejivo_2/aereopuertos_api\$ python manage.py makemigrations

Migrations for 'airports':

- airports/migrations/0001_initial.py
- + Create model Gate
- + Create model Flight

Operations to perform:

- Apply all migrations: admin, airport, auth, contenttypes, sessions

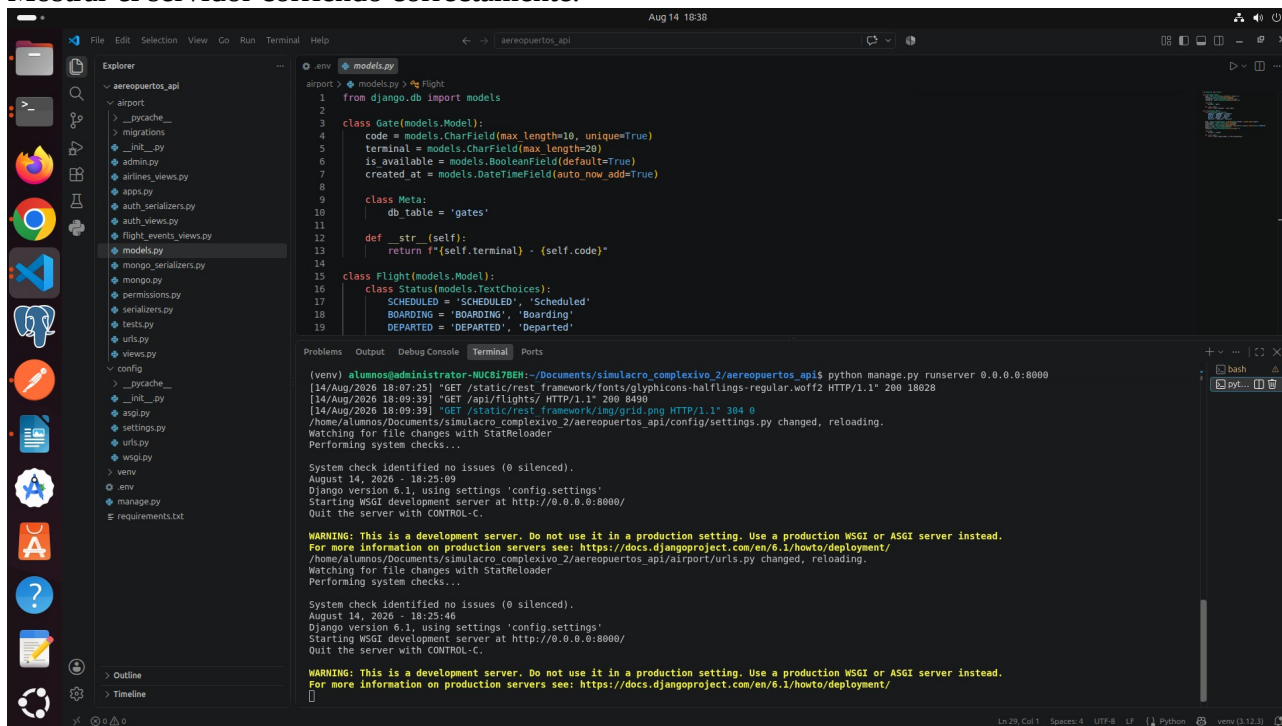
Running migrations:

- Applying contenttypes.0001_initial... OK
- Applying auth.0001_initial... OK
- Applying admin.0001_initial... OK
- Applying admin.0002_logentry_remove_auto_add... OK
- Applying admin.0003_logentry_add_action_flag_choices... OK
- Applying airports.0001_initial... OK
- Applying contenttypes.0002_remove_content_type_name... OK
- Applying auth.0002_alter_permission_name_max_length... OK
- Applying auth.0003_alter_user_email_max_length... OK
- Applying auth.0004_alter_user_username_opts... OK
- Applying auth.0005_alter_user_last_login_null... OK
- Applying auth.0006_require_contenttypes_0002... OK
- Applying auth.0007_alter_validators_add_error_messages... OK
- Applying auth.0008_alter_user_username_max_length... OK
- Applying auth.0009_alter_user_last_name_max_length... OK
- Applying auth.0010_alter_group_name_max_length... OK
- Applying auth.0011_update_proxy_permissions... OK
- Applying auth.0012_alter_user_first_name_max_length... OK
- Applying sessions.0001_initial... OK

(venv) alumnos@administrator-NUC817BEN:~/Documents/simulacro_complejivo_2/aereopuertos_api\$

Captura 17 – Servidor en Ejecución

Mostrar el servidor corriendo correctamente.



```

1 from django.db import models
2
3 class Gate(models.Model):
4     code = models.CharField(max_length=10, unique=True)
5     terminal = models.CharField(max_length=20)
6     is_available = models.BooleanField(default=True)
7     created_at = models.DateTimeField(auto_now_add=True)
8
9     class Meta:
10         db_table = 'gates'
11
12     def __str__(self):
13         return f'{self.terminal} - {self.code}'
14
15 class Flight(models.Model):
16     class Status(models.TextChoices):
17         SCHEDULED = 'SCHEDULED', 'Scheduled'
18         BOARDING = 'BOARDING', 'Boarding'
19         DEPARTED = 'DEPARTED', 'Departed'

```

(venv) alumnos@administrator-NUC817BEN:~/Documents/simulacro_complejivo_2/aereopuertos_api\$ python manage.py runserver 0.0.0.0:8000

[14/Aug/2026 18:07:25] "GET /static/rest_framework/fonts/glyphicons-halflings-regular.woff2 HTTP/1.1" 200 18028

[14/Aug/2026 18:09:39] "GET /api/flights/ HTTP/1.1" 200 8490

[14/Aug/2026 18:09:39] "GET /static/rest_framework/img/grid.png HTTP/1.1" 304 0

/home/alumnos/Documents/simulacro_complejivo_2/aereopuertos_api/config/settings.py changed, reloading.

Watching for file changes with StatReloader

Performing system checks...

System check identified no issues (0 silenced).

August 14, 2026 - 18:25:09

Django version 6.1, using settings 'config.settings'

Starting WSGI development server at http://0.0.0.0:8000/

Quit the server with CONTROL-C.

WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI or ASGI server instead.

For more information on production servers see: <https://docs.djangoproject.com/en/6.1/howto/deployment/>

/home/alumnos/Documents/simulacro_complejivo_2/aereopuertos_api/airports/urls.py changed, reloading.

Watching for file changes with StatReloader

Performing system checks...

System check identified no issues (0 silenced).

August 14, 2026 - 18:25:46

Django version 6.1, using settings 'config.settings'

Starting WSGI development server at http://0.0.0.0:8000/

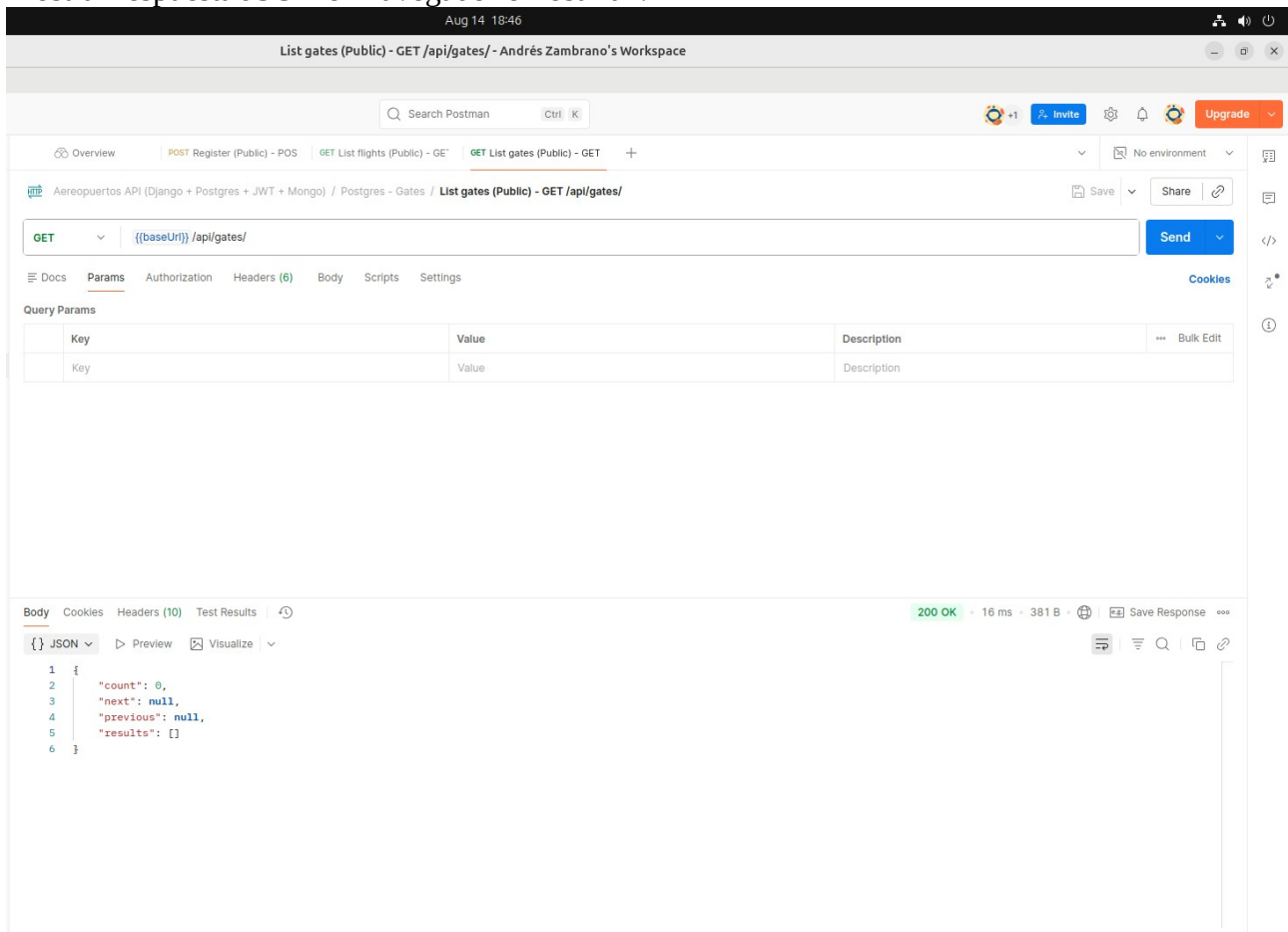
Quit the server with CONTROL-C.

WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI or ASGI server instead.

For more information on production servers see: <https://docs.djangoproject.com/en/6.1/howto/deployment/>

Captura 18 – Endpoint GET Funcional

Mostrar respuesta JSON en navegador o Postman.



Captura 19 – Endpoint POST Funcional

Mostrar inserción exitosa de un registro.

Aug 14 18:47

Register (Public) - POST /api/auth/register/ - Andrés Zambrano's Workspace

Search Postman Ctrl K

Overview POST Register (Public) - POS GET List flights (Public) - GE GET List gates (Public) - GET POST Create gate (Admin) - POST Login (JWT) - POST /a + No environment

Aeropuertos API (Django + Postgres + JWT + Mongo) / Auth / Register (Public) - POST /api/auth/register/ Save Share

POST {{baseUrl}}/api/auth/register/ Send

Docs Params Authorization Headers (9) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```

1 {
2   "username": "user1",
3   "email": "user1@mail.com",
4   "password": "123456"
5 }

```

Body Cookies Headers (10) Test Results

201 Created 308 ms 375 B Save Response

JSON Preview Visualize

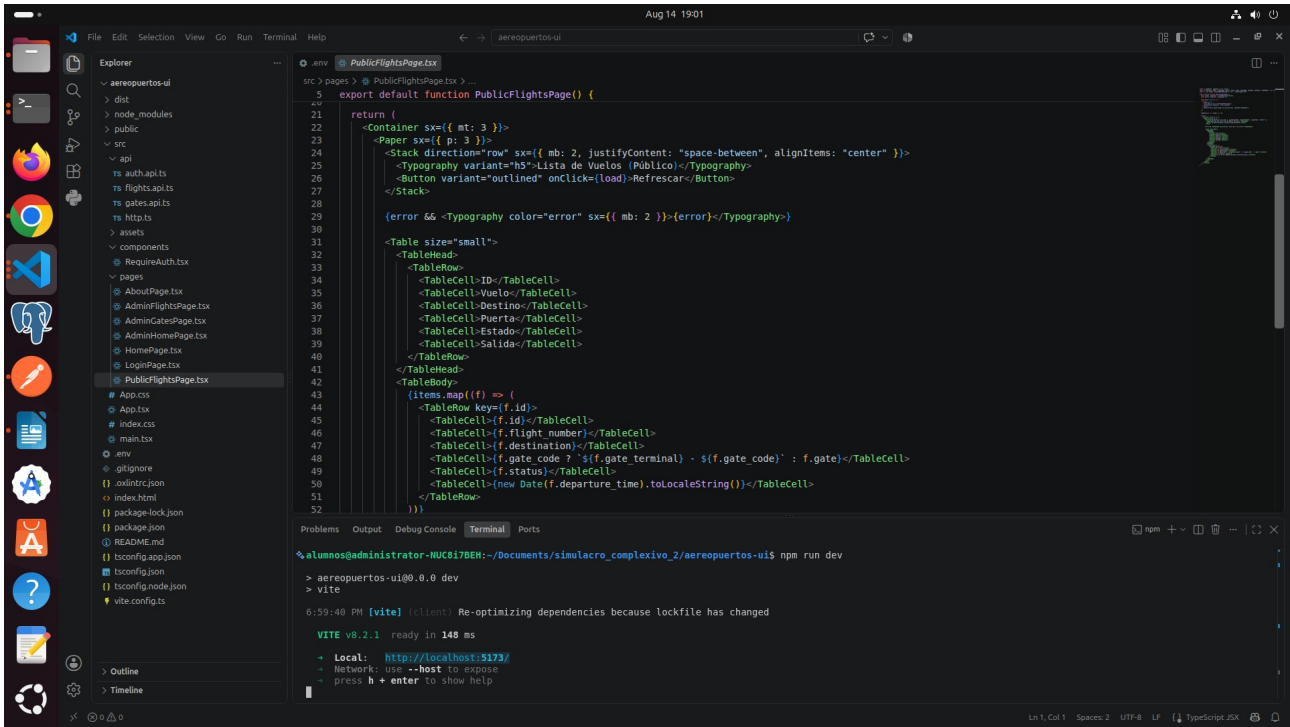
```

1 {
2   "id": 1,
3   "username": "user1",
4   "email": "user1@mail.com"
5 }

```


SECCIÓN 4: FRONTEND – REACTJS

Captura 20 – Proyecto React en Ejecución Mostrar terminal con servidor activo.



The screenshot shows the VS Code editor with the file `PublicFlightsPage.tsx` open. The code defines a `PublicFlightsPage` component that uses `Container`, `Paper`, `Stack`, `Typography`, `Button`, `Table`, and `TableRow` from Material-UI. It includes a loading spinner and a table of flight data. The terminal at the bottom shows the command `npm run dev` being executed, and the output indicates that the development server is running on `http://localhost:5173/`.

```

src > pages > @ PublicFlightsPage.tsx > ...
5 export default function PublicFlightsPage() {
20
21   return (
22     <Container sx={{ mt: 3 }}>
23       <Paper sx={{ p: 3 }}>
24         <Stack direction="row" sx={{ mb: 2, justifyContent: "space-between", alignItems: "center" }}>
25           <Typography variant="h5">Lista de Vuelos (Público)</Typography>
26           <Button variant="outlined" onClick={load}>Refrescar</Button>
27         </Stack>
28
29         {error && <Typography color="error" sx={{ mb: 2 }}>{error}</Typography>}
30
31         <Table size="small">
32           <TableHead>
33             <TableRow>
34               <TableCell>ID</TableCell>
35               <TableCell>Vuelo</TableCell>
36               <TableCell>Destino</TableCell>
37               <TableCell>Puerta</TableCell>
38               <TableCell>Estado</TableCell>
39               <TableCell>Salida</TableCell>
40             </TableRow>
41           </TableHead>
42           <TableBody>
43             {items.map((f) => (
44               <TableRow key={f.id}>
45                 <TableCell>{f.id}</TableCell>
46                 <TableCell>{f.flight_number}</TableCell>
47                 <TableCell>{f.destination}</TableCell>
48                 <TableCell>{f.gate_code ? `${f.gate_terminal} - ${f.gate_code}` : f.gate}</TableCell>
49                 <TableCell>{f.status}</TableCell>
50                 <TableCell>{new Date(f.departure_time).toLocaleString()}</TableCell>
51               </TableRow>
52             ))}
53           </TableBody>
54         </Table>
55       </Paper>
56     </Container>
57   )
58 }

```

```

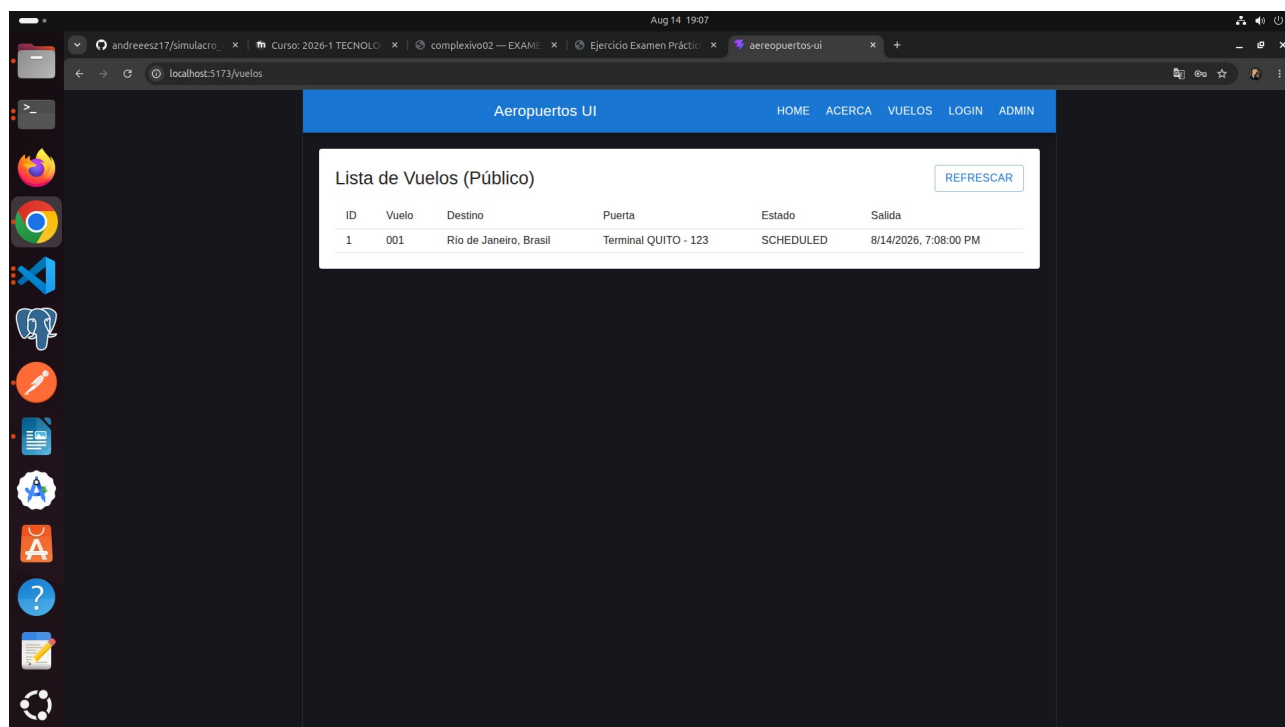
%alumnos@administrator-NUC817BEH:~/Documents/simulacro_complexivo_2/aereopuertos-ui$ npm run dev
> aereopuertos-ui@0.0.0 dev
> vite

6:59:40 PM [vite] (client) Re-optimizing dependencies because lockfile has changed

VITE v8.2.1 ready in 148 ms
  + Local:   http://localhost:5173/
  + Network: use --host to expose
  + press h + enter to show help

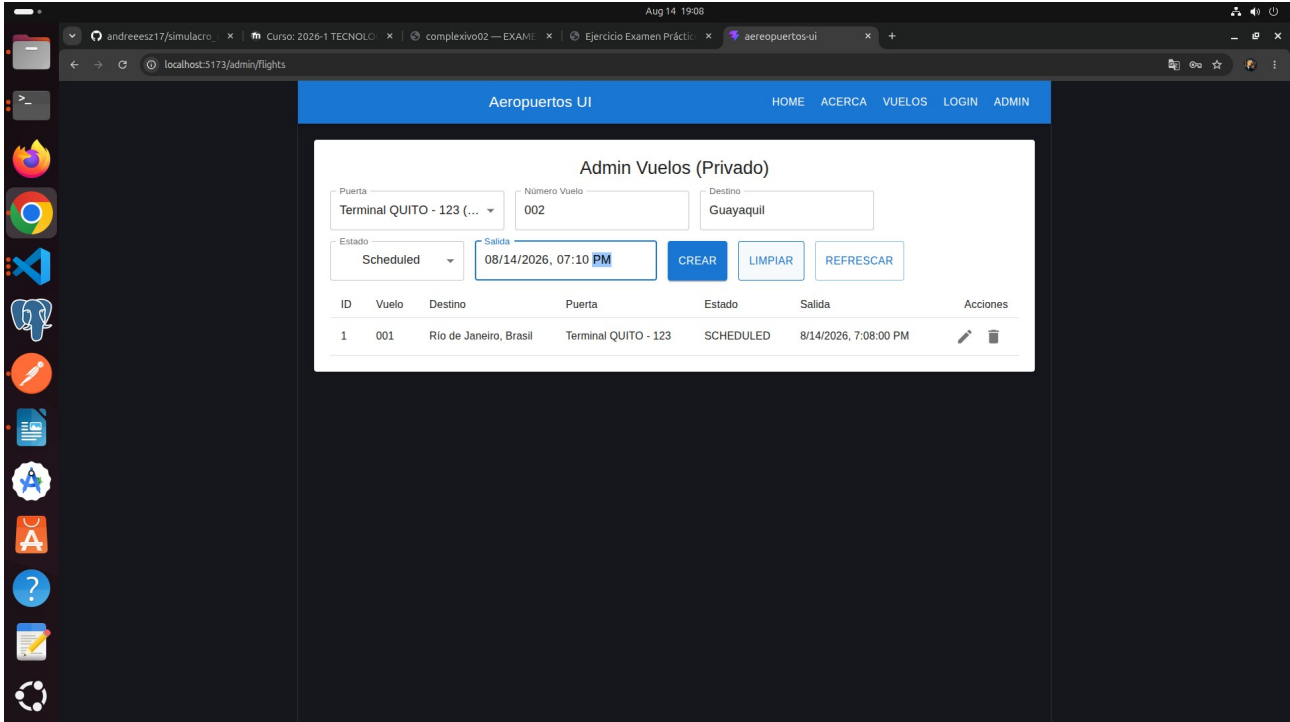
```

Captura 21 – Listado de Registros Mostrar en navegador la lista obtenida desde el backend.



Captura 22 – Registro Nuevo Desde la Interfaz

Mostrar creación de un nuevo registro y actualización del listado.



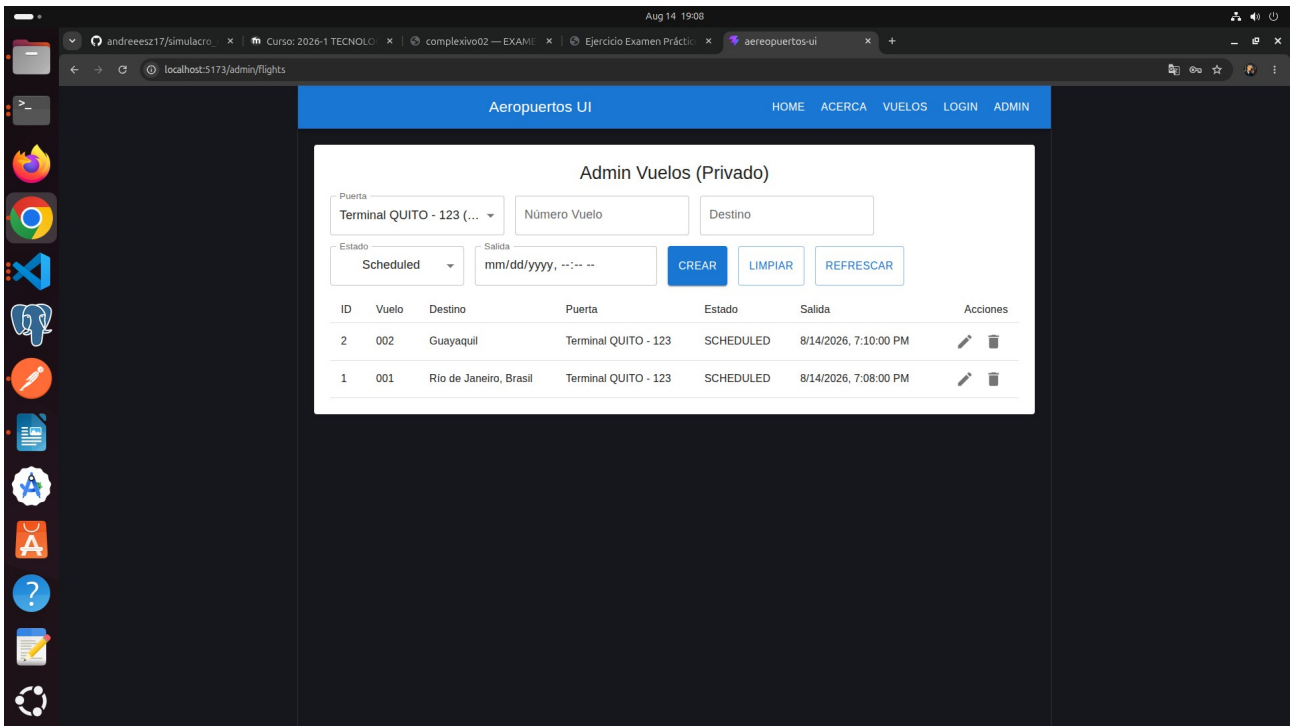
Aeropuertos UI HOME ACERCA VUELOS LOGIN ADMIN

Admin Vuelos (Privado)

Puerta: Terminal QUITO - 123 (...) Número Vuelo: 002 Destino: Guayaquil

Estado: Scheduled Salida: 08/14/2026, 07:10 PM [CREAR] [LIMPIAR] [REFRESCAR]

ID	Vuelo	Destino	Puerta	Estado	Salida	Acciones
1	001	Rio de Janeiro, Brasil	Terminal QUITO - 123	SCHEDULED	8/14/2026, 7:08:00 PM	[Edit] [Delete]



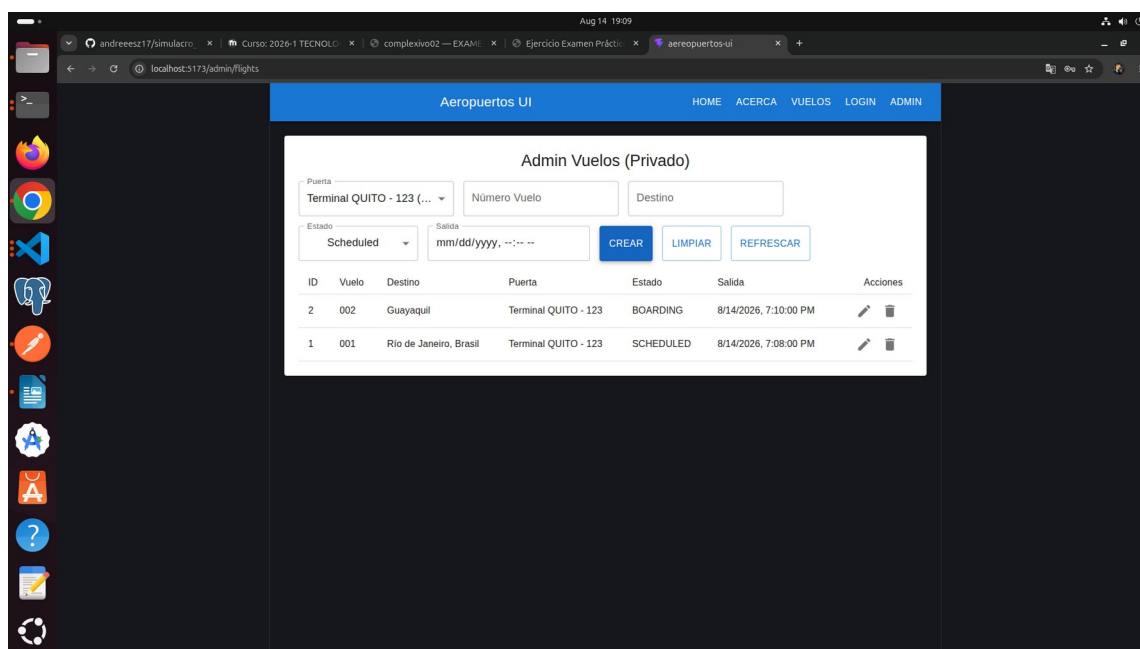
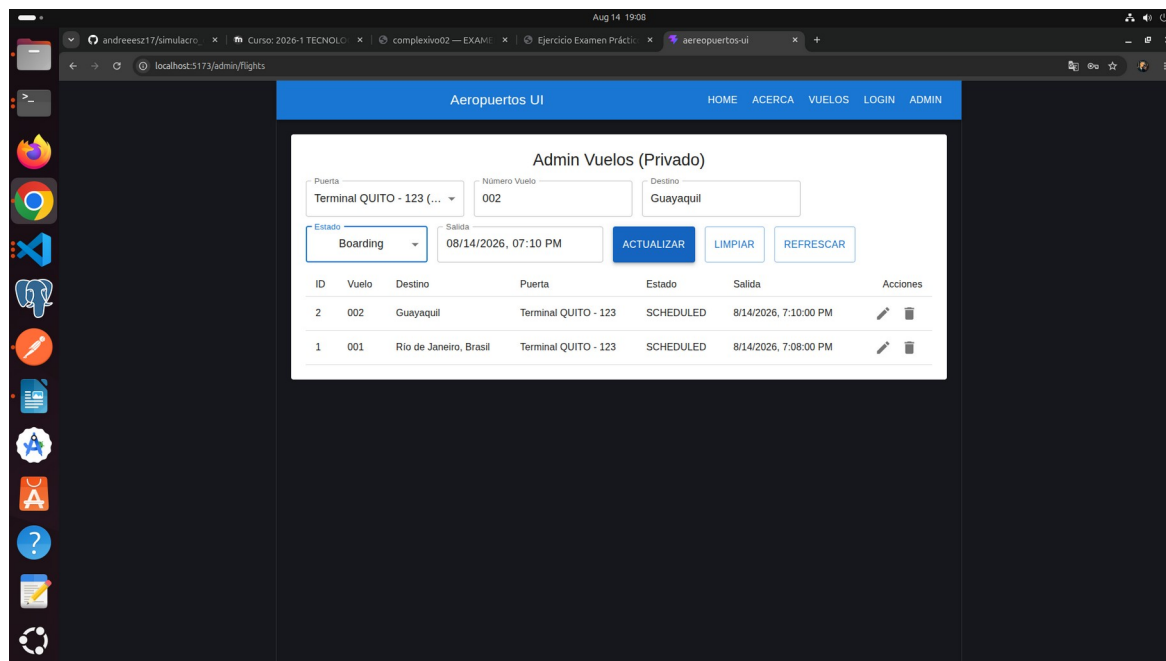
Aeropuertos UI HOME ACERCA VUELOS LOGIN ADMIN

Admin Vuelos (Privado)

Puerta: Terminal QUITO - 123 (...) Número Vuelo: Destino:

Estado: Scheduled Salida: mm/dd/yyyy, --:-- -- [CREAR] [LIMPIAR] [REFRESCAR]

ID	Vuelo	Destino	Puerta	Estado	Salida	Acciones
2	002	Guayaquil	Terminal QUITO - 123	SCHEDULED	8/14/2026, 7:10:00 PM	[Edit] [Delete]
1	001	Rio de Janeiro, Brasil	Terminal QUITO - 123	SCHEDULED	8/14/2026, 7:08:00 PM	[Edit] [Delete]



SECCIÓN 5: APLICACIÓN MÓVIL – REACT NATIVE

Captura 23 – Proyecto Móvil Creado
Mostrar terminal con proyecto inicializado.

Captura 24 – Aplicación Ejecutándose

Mostrar pantalla principal en emulador o dispositivo.

Captura 25 – Consumo de API NoSQL

Mostrar datos provenientes de las colecciones no relacionales.

SECCIÓN 6: SISTEMAS OPERATIVOS – UBUNTU (MÁQUINA VIRTUAL)

Captura 26 – Creación de Estructura de Directorios

Mostrar comandos mkdir y verificación con tree.

```
Aug 14 19:33
alumnos@administrator-NUC8i7BEH: ~/Documents/simulacro_complexivo_2/aereopuerto

alumnos@administrator-NUC8i7BEH:~$ cd D
Descargas/ Desktop/ Documents/ Downloads/
alumnos@administrator-NUC8i7BEH:~$ cd Documents/
alumnos@administrator-NUC8i7BEH:~/Documents$ cd simulacro_complexivo_2/
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2$ mkdir aereopuerto
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2$ cd aereopuerto
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto$ mkdir backend
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto$ mkdir frontend
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto$ mkdir movil
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto$ mkdir docs
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto$ -p
-p: command not found
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto$ tree
.
├── backend
├── docs
├── frontend
└── movil

alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto$
```

Captura 27 – Navegación y Listado

Mostrar uso de cd, pwd y ls -la.

```
Aug 14 19:34
alumnos@administrator-NUC8i7BEH: ~/Documents/simulacro_complexivo_2/aereopuerto

alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto$ pwd
/home/alumnos/Documents/simulacro_complexivo_2/aereopuerto
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto$ ls -la
total 24
drwxrwxr-x 6 alumnos alumnos 4096 Aug 14 19:32 .
drwxrwxr-x 6 alumnos alumnos 4096 Aug 14 19:31 ..
drwxrwxr-x 2 alumnos alumnos 4096 Aug 14 19:31 backend
drwxrwxr-x 2 alumnos alumnos 4096 Aug 14 19:32 docs
drwxrwxr-x 2 alumnos alumnos 4096 Aug 14 19:32 frontend
drwxrwxr-x 2 alumnos alumnos 4096 Aug 14 19:32 movil
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto$
```

Captura 28 – Redirección de Salida

Mostrar uso de > y >> con comandos básicos.

```
Aug 14 19:40
alumnos@administrator-NUC8i7BEH: ~/Documents/simulacro_complexivo_2/aereopuerto/docs
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto/docs$ cd docs
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto/docs$ touch aereo
puerto_comandos.txt evidencia.txt
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto/docs$ date >> evi
dencia.txt
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto/docs$ cat evidenc
ia.txt
Fri Aug 14 07:38:27 PM -05 2026
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto/docs$ who > usuar
ios.txt
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto/docs$ ls -la >> u
suarios.txt
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto/docs$ cat usuario
s.txt
alumnos  seat0      2026-08-14 17:07 (login screen)
alumnos  tty2      2026-08-14 17:07 (tty2)
alumnos  pts/11    2026-08-14 18:04
total 16
drwxrwxr-x 2 alumnos alumnos 4096 Aug 14 19:39 .
drwxrwxr-x 6 alumnos alumnos 4096 Aug 14 19:32 ..
-rw-rw-r-- 1 alumnos alumnos  0 Aug 14 19:37 aereopuerto_comandos.txt
-rw-rw-r-- 1 alumnos alumnos  32 Aug 14 19:38 evidencia.txt
-rw-rw-r-- 1 alumnos alumnos 139 Aug 14 19:39 usuarios.txt
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto/docs$
```

Captura 29 – Búsqueda con grep

Mostrar búsqueda de texto dentro de un archivo.

```

Aug 14 19:45
alumnos@administrator-NUC8i7BEH: ~/Documents/simulacro_complexivo_2/aereopuerto/docs
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto/docs$ cat << EOF
> aereopuerto_comandos.txt
Proyecto Aeropuerto - Backend
GET /api/flights/
GET /api/flights/15/
POST /api/flights/
DELETE /api/flights/10/
INFO: flight created successfully
INFO: flights service running
WARN: flight delay detected
EOF
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto/docs$ cat aereopu
erto_comandos.txt
Proyecto Aeropuerto - Backend
GET /api/flights/
GET /api/flights/15/
POST /api/flights/
DELETE /api/flights/10/
INFO: flight created successfully
INFO: flights service running
WARN: flight delay detected
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto/docs$ grep -n -i
"flights" aereopuerto_comandos.txt
grep: aereopuerto_comandos.txt: No such file or directory
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto/docs$ grep -n -i
"flights" aereopuerto_comandos.txt
2:GET /api/flights/
3:GET /api/flights/15/
4:POST /api/flights/
5:DELETE /api/flights/10/
7:INFO: flights service running
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto/docs$

```

Captura 30 – Permisos y Sticky Bit

Mostrar uso de chmod y verificación de permisos con ls -ld.

```

Aug 14 19:49
alumnos@administrator-NUC8i7BEH: ~/Documents/simulacro_complexivo_2/aereopuerto
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto/docs$ cd ..
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto$ mkdir shared_aer
eopuerto
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto$ chmod 1777 share
d_aereopuerto
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto$ ls -ld shared_ae
reopuerto
drwxrwxrwt 2 alumnos alumnos 4096 Aug 14 19:48 shared_aereopuerto
alumnos@administrator-NUC8i7BEH:~/Documents/simulacro_complexivo_2/aereopuerto$

```